

KEY BATTLE

CHARGE OF THE
LIGHT BRIGADE
25 OCTOBER 1854

During the Crimean War, Lord Cardigan led an attack by British hussars, lancers, and dragoons on a heavy Russian artillery battery. Charging along a valley covered by dense Russian fire, the light cavalry lost almost half their number.



▲ The British horsemen reached the Russian guns, but at the cost of an excessive number of lives.

KEY DEVELOPMENT

THE BEGINNINGS OF
MODERN WARFARE

In the 19th century, weaponry was transformed by the cumulative effect of technological innovations. Armies clung to well-established tactics, such as the cavalry charge and frontal infantry assault, but improved firepower made these methods increasingly obsolete.

The transformation of infantry weapons began in 1839, with the adoption of the percussion cap as an ignition system for muskets, over the long-established flintlock mechanism. However, in the 1840s, the Prussian army rearmed its infantry with the Dreyse needle gun, a bolt-action, breech-loading rifle firing a cartridge with integrated percussion cap, powder, and bullet. This had a higher rate of fire than muskets, and could be fired lying or kneeling behind cover.

In 1853, the British Army adopted the Enfield rifled musket, which fired the Minié ball: this ingenious bullet was small enough to be rammed down the barrel, but expanded when fired to engage the rifling. First used in the Crimean War (1854–56), fought between Russian and allied forces, the rifled musket had more than twice the range of

a conventional musket, and it became the standard infantry weapon of the American Civil War (1861–65). It greatly increased the infantry casualties across open ground, even when they advanced in a loose skirmish line, rather than traditional columns.

TECHNOLOGICAL DEVELOPMENTS

In the Franco-Prussian War of 1870–71, infantry on both sides used breech-loading rifles firing the self-contained cartridge, an invention that led to repeater rifles, revolvers, and machine-guns: early rapid-fire weapons emerged in the 1860s, with the French Mitrailleuse and the American Gatling gun (see pp.246–47), although the first true machine-gun, the Maxim gun, came into service in the 1880s. Cavalry in the American Civil War often carried Colt or Remington revolvers and Spencer repeating carbines, but only a few infantry had repeating rifles.

By the 1880s, advanced armies were adopting bolt-operated magazine rifles as standard, whereas innovations in artillery were slower. The American Civil War featured muzzle-loading, smoothbore cannon that were only a marginal improvement on

▲ THE COLT
DRAGOON REVOLVER

Supplied to American mounted troops from 1848, the Colt Dragoon pistol had six revolving chambers. The soldier loaded each chamber with a ball, black powder, and a percussion cap.

“The fire was so destructive my line wavered like a man trying to walk against [the] wind”

CONFEDERATE COLONEL WILLIAM OATES, WRITING ABOUT GETTYSBURG, 1905

